

22. **Ans: A**

Going through crossed fields undeviated: $F_E = F_B$

$$qE = B_1 qv$$

$$\text{Speed of particle, } v = \frac{E}{B_1}$$

Going through 2nd field, undergo circular motion of diameter 60 cm:

$$\frac{mv^2}{r} = B_2 qv$$

$$m = \frac{B_2 qr}{v}$$

$$= \frac{B_1 B_2 qr}{E}$$

$$= \frac{2.0 \times 10^{-5} \times 1.2 \times 2 \times 1.60 \times 10^{-19} \times 0.30}{6.0 \times 10^2}$$

$$= 3.8 \times 10^{-27} \text{ kg}$$

23. **Ans: C**